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- This exam contains 11 pages (including this cover page) and 17 problems. Check to see if any pages are missing.
 - This is the second exam and will count for 20% of your final grade. You have 115 minutes (1:35 pm - 3:30 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work when work is required will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 300 points total in this exam, as well as 3 bonus questions worth a total of 60 possible points.

Page:	3	4	5	6	7	8	Total
Points:	80	40	60	50	20	50	300
Score:							

1. 20 points Find $C'(x)$ given that the total cost (in hundreds of dollars) of producing x speed-boat hulls is

$$C(x) = 50 + 2(4x^3 + 2)^{-3} + 6x^2.$$

work

2. For the following problems, find $\frac{dy}{dx}$.

(a) 20 points $2x^3 = 9y^3 + 120x$

work

(b) 20 points $\frac{1 + 2x^2}{y^2 - 1} = x^2$

work

(c) 20 points $x \ln(x^5 + x + 1) = e^{3y^3 + 3}$

work

3. 20 points If $f(x) = e^{(x+9)^{100}}$, find the value(s) of x where the tangent line of f is horizontal.

work

4. Let $x = x(\star)$, $t = t(\star)$ (that is, if both x and t represent functions of $\star$), and suppose that x and t are related by

$$2e^{x^3+x-1} = e^4 + (xt)^2.$$

- (a) 10 points Find $\frac{dt}{d\star}$

work

- (b) 10 points Find $\frac{dt}{d\star}$ when $x = 2$, $t = 1$, and $\frac{dx}{d\star} = 7$.

work

5. 20 points Find the relative rate of change of $f(x) = 3 + e^x(x - 5)^9$.

work

6. 20 points Given the price-demand equation $x = f(p) = p + (1 - p^2)^{16}$, find $E(p)$, the price-elasticity of demand. At $p = 0.5$, is demand elastic or inelastic?

work

7. 20 points Given the price-demand equation $x = f(p) = e^{100-p^3}$, find the value(s) of the price p for which demand has unit elasticity.

work

8. 20 points Given the price-demand equation $x = f(p) = 10\ln(p^2 + 1)$, what price p **minimizes** the revenue? (*Hint: $R(p) = pf(p)$ and $\ln(1) = 0$*)

work

9. 10 points If $f(x) = \frac{x^{-1}}{2^{-1}} - \frac{x^{-2}}{2^{-1}}$, find the intervals on which $f(x)$ is increasing.

work

10. 20 points Let $f(x) = (x-9)^5 e^{-5x}$. Find the intervals on which $f(x)$ is increasing, the intervals on which $f(x)$ is decreasing, and the local extrema (if there are any).

(*Hint: factor by grouping to find all the roots of your derivative.*)

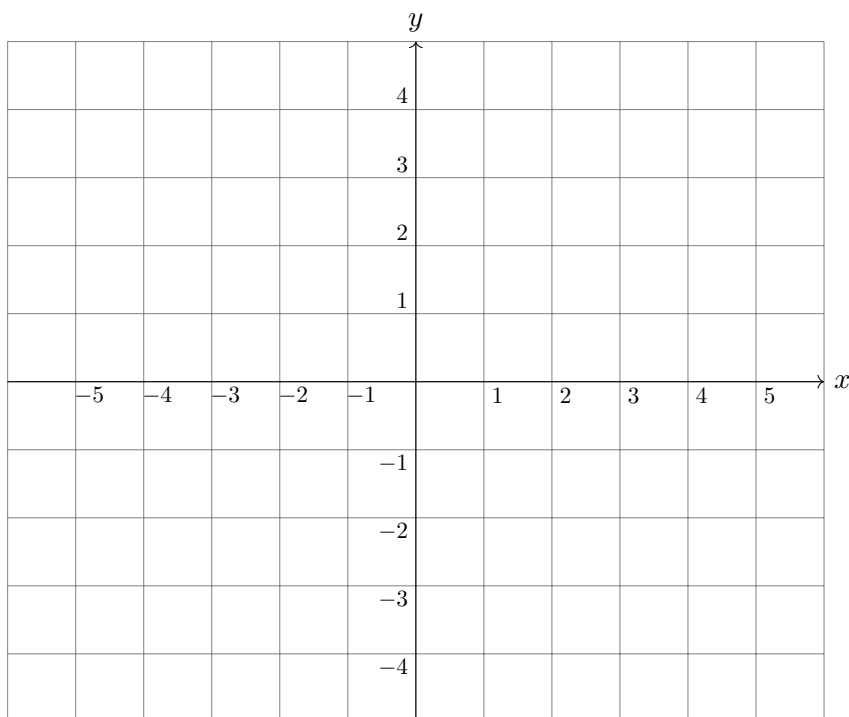
work

11. 20 points Sketch a graph of a function $f(x)$ which is continuous on $(-\infty, \infty)$ and satisfies:

$$f(3) = 4, \quad f'(-1) = 0, \quad f'(3) = 0, \quad f''(1) = 0, \quad f''(-1) = 0$$

$$f'(x) > 0 \text{ on } (-1, 3) \text{ and } (-\infty, -1), \quad f'(x) < 0 \text{ on } (3, \infty).$$

(*Hint*: think about x^3)



12. 10 points If $y = e^{3x} + (x - 1)^3$, find y'' .

work

13. 20 points If $f(x) = e^{3x} - 3e^{\sqrt{3}x} + x^3$, find any values of x for which $f''(x) = 0$.
(*Hint:* $e^0 = 1$)

work

14. 20 points If $f(x) = 27x^2 - 9x^{-2}$, find the x and y coordinates of all inflection points.

work

15. 20 points (bonus) Let y represent a function of x . Derive a general formula for $\frac{d}{dx} (x^n e^{y^m})$. Show all of your steps.

work

16. 20 points (bonus) A farmer estimates that yearly crop sales s and yearly seed costs c (both in dollars) are related by

$$s = 4,500 + 0.2(c^2 - 90)^4.$$

The current yearly seed costs are \$20,000, and these are increasing at a rate of \$150 per year. Find the current rate of change in yearly crop sales sales.

work

17. 20 points (bonus) Given the price-demand equation $x = f(p) = 200 - p^2$, find the equation of the line tangent to the graph of the elasticity of demand function at $p = 2$.

work